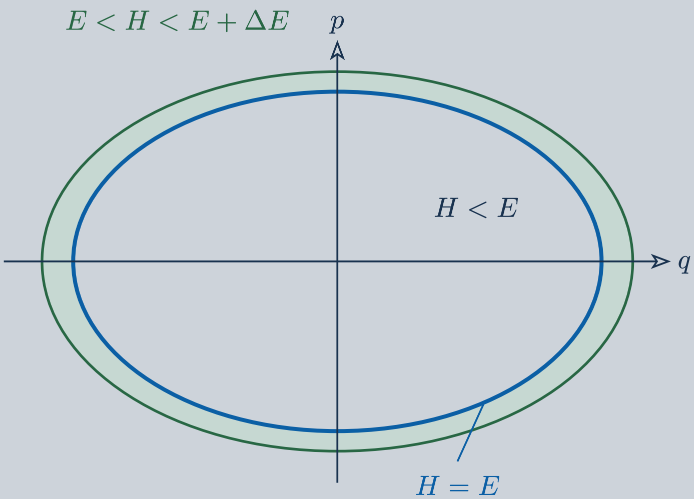
1 One oscillator: volume, contour, shell



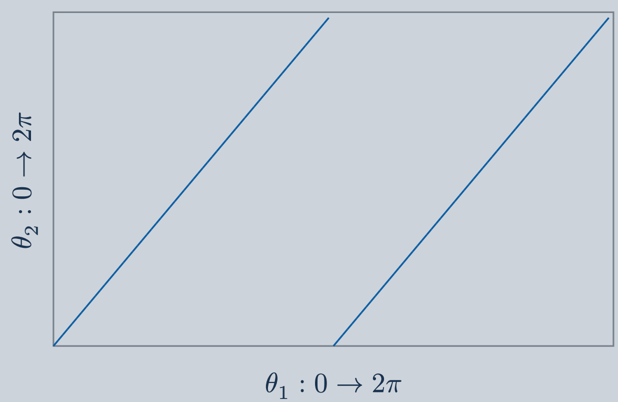
$H(q,p) = \frac{p^2}{2m} + k\frac{q^2}{2}$

$q_{\max} = \sqrt{2\frac{E}{k}}, \quad p_{\max} = \sqrt{2mE}$

Interior: $H < E$
Blue contour: $H = E$
Green shell: $E < H < E + \Delta E$

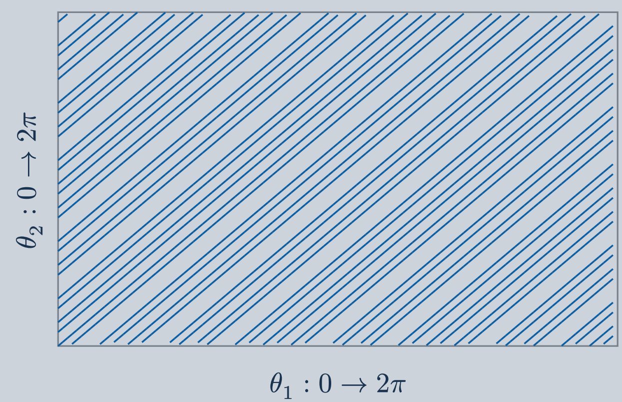
Axes scaled independently; shell exaggerated.

2 Rational: closed orbit



$\omega_2/\omega_1 = 2 \cdot$ Opposite edges identified.

3 Irrational: dense orbit



$\omega_2/\omega_1 = \sqrt{2} \cdot$ Finite segment shown.

Irrational flow is ergodic on its **fixed-action torus** with uniform angle measure, not generally on the full energy surface.